

Azure Data Factory Project

1. Pipeline

Requirements of pipeline:

- Copy all the csv sales data files from source container to sink container
- In sink container for all files should go in hierarchy folder like year/month.csv

1. File Setup in Blob Storage

Place all the files in the source-container.

Microsoft Azure

Home > blobstorageproject1 | Containers >

source-container Container

Search

Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot Give feedback

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: source-container

Search blobs by prefix (case-sensitive) Show deleted blobs

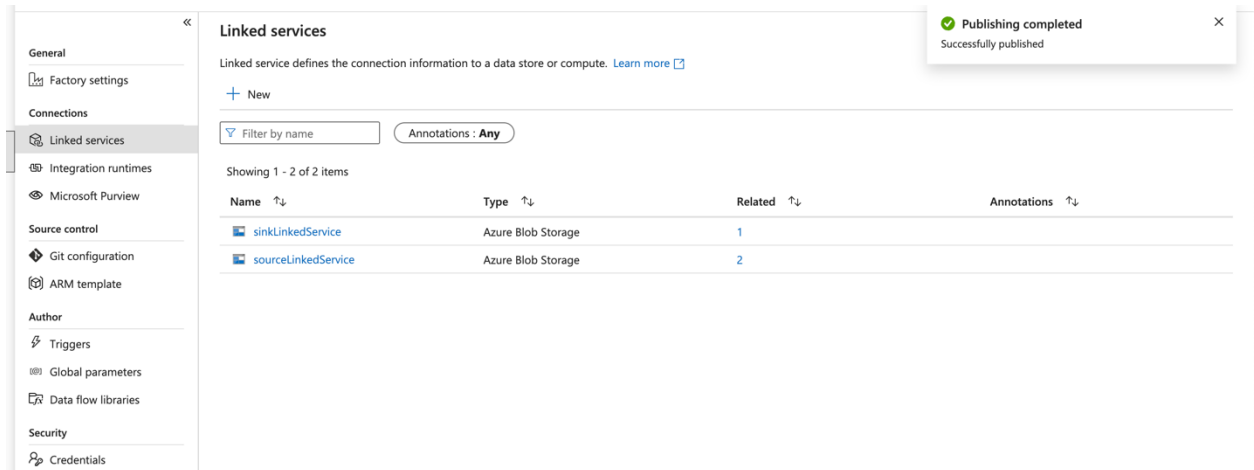
Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
☐ 2020-04.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.44 KiB	Available ...
☐ 2020-05.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.72 KiB	Available ...
☐ 2020-06.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.28 KiB	Available ...
☐ 2020-07.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.54 KiB	Available ...
☐ 2020-08.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.79 KiB	Available ...
☐ 2020-09.csv	1/5/2025, 5:31:57 PM	Hot (Inferred)		Block blob	7.71 KiB	Available ...
☐ Data Engineering Roadmap-Ashish.xlsx	1/5/2025, 7:03:12 PM	Hot (Inferred)		Block blob	8.55 KiB	Available ...

- Create a sink container

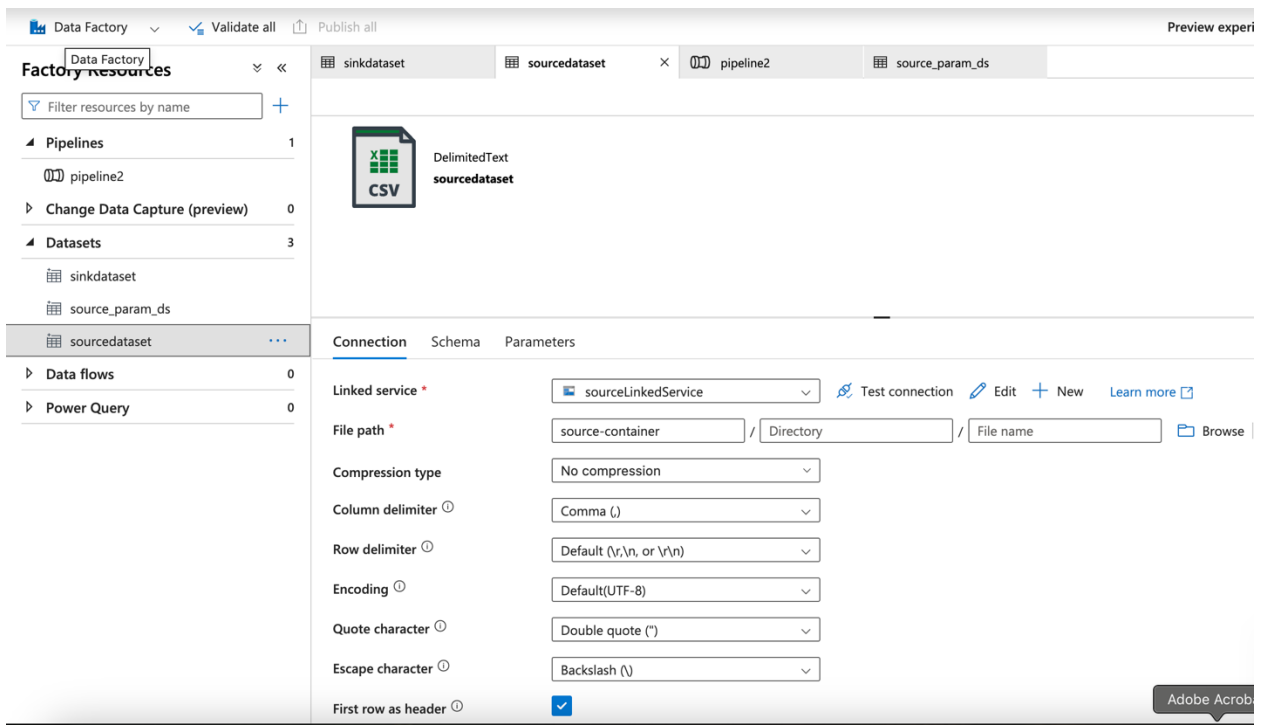
2. Setup Azure data factory instance

2.1 create a linked service



2.2 create a datasets

Source dataset



sink dataset

The screenshot shows the configuration for a sink dataset named 'sinkdataset'. The left sidebar lists the hierarchy: pipeline2 > Datasets > sinkdataset. The main panel has tabs for Connection, Schema, and Parameters. The Parameters tab is active, showing the following settings:

- Linked service: sinkLinkedService
- File path: sink-container / @dataset().year / @dataset().month
- Compression type: No compression
- Column delimiter: Comma (,)
- Row delimiter: Default (\r\n, or \r\n)
- Encoding: Default(UTF-8)
- Quote character: Double quote (")

The screenshot shows the 'Factory Resources' sidebar on the left, listing Pipelines (pipeline2) and Datasets (sinkdataset, source_param_ds, sourcedataset). The main panel shows the 'Parameters' tab for the 'sinkdataset'.

Parameters Table:

Name	Type	Default value
year	String	Value
month	String	Value

2.3 create a pipeline

The screenshot shows the 'pipeline2' configuration interface. The left sidebar lists 'Activities' such as Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, and Databricks. The main panel shows the pipeline flow:

```
graph LR; GetMetadata[Get Metadata] --> Filter[Filter1]; Filter --> ForEach[ForEach1]; subgraph ForEach1 [ForEach1] direction TB; Activities[Activities] --> CopyData1[Copy data1]; end
```

Pipeline Activities

Activity 1: Get Metadata

✓ Validate

▶ Debug

⚡ Add trigger

Get Metadata

Get Metadata1

🗑

</>

📄

➔

✓

✗

➔

Filter

Filter1

ForEach

Activities

Copy data1

General

Settings

User properties

Dataset *

sourcedataset

Open

New

Field list *

+ New

🗑 Delete

☐ Argument

☐ Child items

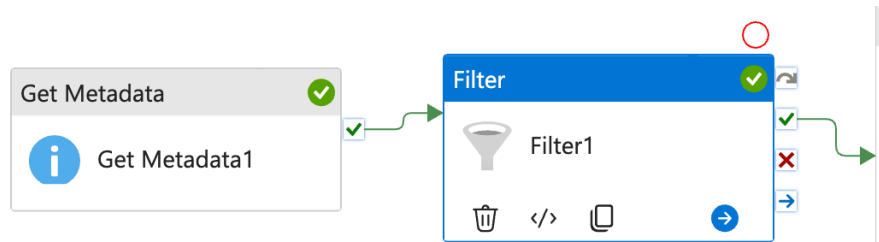
Filter by last modified ⓘ

Start time (UTC)

End time (UTC)

Skip line count

Activity 2 filter get all the csv files



General **Settings** User properties

Items

@activity('Get Metadata1').output.chi...

Condition

@contains(item().name, '.csv')

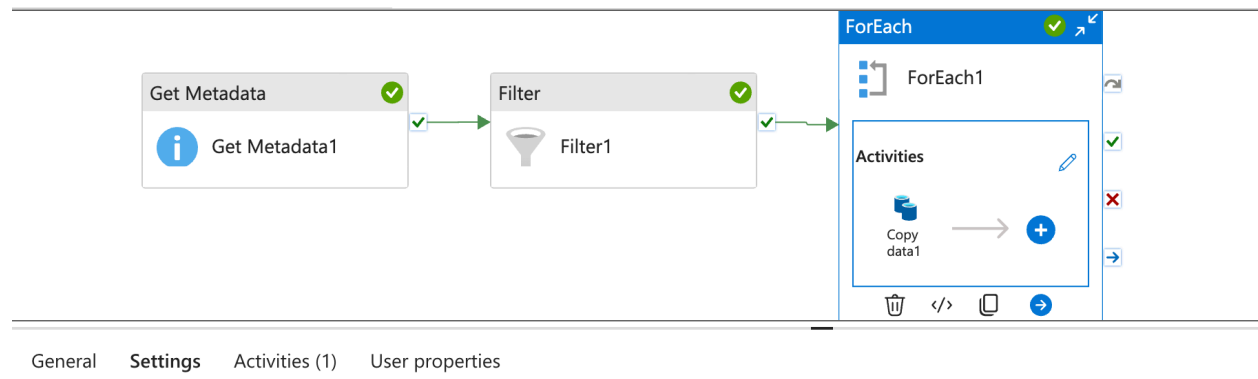
Output

Copy to clipboard

```
{
  "ItemCount": 7,
  "FilteredItemCount": 6,
  "Value": [
    {
      "name": "2020-04.csv",
      "type": "File"
    },
    {
```

Activity 3 use for each to copy data

✓ Validate ▶ Debug ⚡ Add trigger



In copy data activity

Source :

1. Create a Source dataset with parameter.



Connection	Schema	Parameters
Linked service *	source_LS Test connection Edit + New Learn more	
File path *	source-container / Directory	@dataset().Filename Browse
Compression type	No compression	
Column delimiter	Comma (,)	
Row delimiter	Default (\r\n, or \r\n)	
Encoding	Default(UTF-8)	

Connection

Schema

Parameters

+

New

🗑️

Delete

☐	Name	Type	Default value	
☐	Filename	String	Value	🗑️

For sink dataset

🗑️

</>

📄

➡️

⌵

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

sinkdataset

🔗

Open

+

New

📖

Learn more

Dataset properties ⓘ

Name	Value	Type
month	@substring(item().name, 5, 6)	String
year	@substring(item().name, 0, 4)	String

Copy behavior ⓘ

Select...

Validate debug and run:

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✓ Validate

▶️ Debug

⚡ Add trigger

{}

📄

⋮

Get Metadata

Filter

ForEach

Copy data1

sinkdataset

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	15s	AutoResolveIntegrator		e53d2dd4-2f3
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	15s	AutoResolveIntegrator		35cddef6-897
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	14s	AutoResolveIntegrator		6f2acfd7-2d2c
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	17s	AutoResolveIntegrator		cbe8bf5d-564
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	16s	AutoResolveIntegrator		d02a3c48-0e8
Copy data1	✓ Succeeded	Copy data	1/7/2025, 1:47:30 PM	14s	AutoResolveIntegrator		5d373a64-c87
ForEach1	✓ Succeeded	ForEach	1/7/2025, 1:47:29 PM	21s			8d7966e5-fca
Filter1	✓ Succeeded	Filter	1/7/2025, 1:47:28 PM	Less than 1s			605f4c1b-339
Get Metadata1	✓ Succeeded	Get Metadata	1/7/2025, 1:47:21 PM	7s	AutoResolveIntegrator		2a52b560-4fb

Sink container in BLOB:

